MIL-W- 21985(Aer)
13 March 1959

MILITARY SPECIFICATION

WHEELS, - CUSHION TREAD (SEMI - PNEUMATIC)
FOR MOBILE GROUND SUPPORT EQUIPMENT

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification covers one type of cushion tread (semi-pneumatic) wheels for use on mobile ground support equipment.

2. APPLICABLE DOCUMENTS

2.1 General: The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS:

Federal

TT-E-529	Enamel, Synthetic Semi-gloss
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and lock-corner

Military

MIL-B-121	Barrier Material, Greaseproofed flexible (Waterproofed)
MIL-D-5028	Drawings and Data Lists; Preparation of Manufacturers (For Production Aircraft, Guided Missiles, Engines, Accessories and other Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-W-8005	Wheels and Hubs, for Industrial Pneumatic Tires
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-G-10924	Grease, Automotive and Artillery

STANDARDS:

Federal

FED-STD-595	Colors; (For), Ready Mixed Paints
FED-STD-601	Rubber, Sample Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS-24328	Hub-Cantilever Mounting for 6.00-9, 7.50-10, and 9.00-10 Wheels
MS-33586	Metals-Definition of Dissimilar

FSC- 2530

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PUBLICATIONS:

Air Force - Navy Aeronautical Bulletin

143 Specifications and Standards; Use of

(When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Aviation Supply Depot, Philadelphia, Pennsylvania, Attention: Code CDD).

3. REQUIREMENTS:

3.1 Qualifications: The wheels furnished under this specification shall be a product which has been tested and has passed the preproduction tests specified herein.

3.2 Preproduction Testing: If the wheels to be furnished hereunder have not previously qualified as specified above, a preproduction sample shall be furnished by the contractor for inspection and testing as hereinafter specified to determine conformance with this specification. Production of wheels under the contract shall not be started until the preproduction sample has been approved. All testing shall be at the contractor's expense.

3.2.1 Approval: Approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. No changes or deviations from the approved preproduction sample will be acceptable without prior written approval from the technical bureau.

3.2.2 Components: The wheel assembly shall consist of a one piece wheel with a cushion tread rubber tire attached, two (2) taper roller bearings, grease seal and hub cap.

3.3 Selection of Specifications and Standards: Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143.

3.3.1 Standard Parts: MS and AN Standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Meterials:

3.4.1 Protective Treatment: The use of any protective coating that will crack, chip or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Metals: Metals shall be of the corrosion-resistant type and/or shall be suitably protected to resist corrosion during normal service life.

3.4.3 Dissimilar Metals: Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces shall be permitted. Dissimilar metals are defined in MS-33586.

3.5 Design: The wheel shall be used on ground portable equipment which must be towed or moved by hand on paved areas, pierced plank areas, unpaved roads and unimproved level terrain whether such areas are dry, wet, or covered with up to 6 inches of snow or 4 inches of mud.

3.5.1 Construction: The wheel shall be one piece aluminum casting with an integral brake drum. The tire shall be bonded to the wheel.

3.4.1.1 Wheel: The wheel shall be one of two sizes, Model I on which the tread bonding surface is to be 11 inches diameter by 4 inches wide and Model II on which the tread bonding surface is to be 15 inches diameter by 6 inches wide. The integral brake drum shall be of dimensions and located as shown in Figure I. The braking surface shall have a finish of a minimum 250 micro inches RMS.

3.5.1.2 Hub Portion: The hub portion of the wheel casting shall conform with MS-24323 (Specification MIL-W-8005) for interchangeability with cantilever mounting for sizes 6.00-9, 7.50-10 and 9.00-10 pneumatic wheels.

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3.5.1.3 Tire Portion: The tire portion shall have an outer case, one half (½) inch thick, having hardness of 70 ± 3 durometers, and an inner core having a hardness of 55 ± 3 durometers. The tire depth shall be 2½ inches for Model I wheels and three (3) inches for Model II wheels. The overall dimensions for the tires shall be 16 inches in diameter by four (4) inches wide for Model I and 21 inches in diameter by six (6) inches wide for Model II. The tire shall have a tread design as approved by the procuring agency.

3.5.1.4 Weight: The weight of the complete wheel assembly shall not exceed twenty-eight (28) pounds for Model I and sixty-five (65) pounds for Model II.

3.5.1.5 Seals: Seals for the hub as shown on the Standard MA-24328 shall be standard grease seals of the molded synthetic-rubber type. The seals shall provide adequate protection for the wheel bearings under the conditions specified herein.

3.5.1.6 Bearings: Roller bearings for the hub portion shall be provided as shown on Standard MS-24328 and must meet the performance requirements specified herein. The bearings shall be packed with grease conforming to Specification MIL-G-10924 during assembly for shipment. Provisions shall be provided inside the hub portion to facilitate the removal of the bearing cones.

3.5.1.7 Wheel Assembly: The wheel assembly shall, in addition to that specified above, consist of necessary nuts, lock washers, cotter pins, hub cap, etc., necessary for attaching the wheel to an axle as specified in Standard MS-24328.

3.6 Performance: The wheels shall be capable of operation as specified in the towing requirements of SECTION 4 herein and under the rated loads of Table I when subjected to the following environmental conditions:

a. A temperature of from -65° to + 125°F.
b. A relative humidity of 95 ± 5 percent at a temperature of 125° ± 5°F.
c. Exposure to airborne sand and dust particles encountered in normal and desert operation.
d. Exposure to atmosphere containing salt laden moisture.

3.7 Finishes and Protective Coatings:

3.7.1 Exposed Parts and Surfaces: Exterior surfaces shall be treated in accordance with Specification MIL-A-8625 or MIL-C-5541. A primer coat conforming to MIL-P-8585 shall be applied.

3.7.2 Rubber Products: The rubber portion of the wheel shall not be painted.

3.7.3 Finish Coat: The exterior portion of the wheels shall be finished with two coats of enamel, color, yellow in accordance with color 23538 of Specification Federal Standard 595, conforming to Specification TT-E-529.

3.7.4 Interior Surfaces: Sealed or bearing surfaces designed to permit lubrication, brake drum surface, etc., shall not be painted.

3.8 Identification of Product: Wheels shall be marked for identification in accordance with Standard MIL-STD-130. The identification shall be placed so as to be visible when the wheels are installed. The marking shall be plainly visible after protective finishes have been applied and shall consist of the following:

Specification_____
Model_____Size_____
Manufacturer's Name or Trade Mark_____

3.9 Interchangeability: All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part number.

3.10 Workmanship: The wheel assemblies shall be fabricated and finished in accordance with sound commercial practices. Particular attention shall be given to the smoothness of finish and thoroughness of painting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests: The tests shall be classified as follows:

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a. Acceptance Tests
b. Preproduction Sample Testing

4.1.1 Acceptance Tests: The acceptance tests shall consist of the following:

a. Individual Tests
b. Sampling Plan and Tests

4.1.2 Individual Tests: Each wheel shall be subjected to the following tests:

a. Examination of Product (Visual inspection to determine compliance).
b. Compression Set
c. Overload Test
d. Tensile Strength

4.1.3 Sampling Plan and Test: Two (2) wheels shall be selected at random from every hundred or fraction thereof produced and subjected to the following tests:

a. All the individual tests
b. Roll Tests
c. Towing Tests
d. Speed Tests

4.1.4 Rejection and Retest: When one wheel assembly from a production lot fails to meet the specification, acceptance of all wheels in the production run shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.1.4.1 Defects in Wheels Already Accepted: The investigation of a test failure could indicate that defects may exist in wheels already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and method of correcting or returning of the items.

4.2 Test Conditions:

4.2.1 Tolerances: The capacities, pressures, temperatures and other data required for testing specified herein shall be accurate to within a tolerance of ± 2 percent. Data during endurance, environmental and individual inspection tests shall be accurate to within a tolerance of ± 4 percent.

4.2.2 Marking of Test Sample: All wheels submitted for preproduction testing shall be plainly identified by securely attached double tags marked with the following information:

Sample Wheel: Model_____Size_____
Contract No._____
Manufacturer's Name_____
Manufacturer's Part No._____

If wheels are submitted for qualification and no contract exists between the manufacturer and the Government the following shall be added to the above:

Specification No._____
Authority to Submit_____

4.3 Preproduction Tests:

4.3.1 Radial Load Test: A radial load equal to five (5) times the rated load specified in Table I for the applicable wheel size shall be applied through a flat steel plate to a wheel resting on a flat steel plate in such a manner that the direction of loading is parallel to the plane of the wheel and through the centerline of the hub. The load shall be made to act upon the wheel for a period of one (1) minute. After the load is removed, the wheel shall be inspected for failure or deformation. Deformation shall be grounds for rejection.

4.3.2 Side Load Test: The side load shall be applied to either side of the wheel and in a direction paralled to the axis of the hub by means of a block to the side of the tire. The load shall be applied uniformly to the block which shall conform to the contour of the side of the tire, covering an arc of not more than 60 degrees and whose centroid falls on a point midway between the rim flange outside

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diameter and the tire inside diameter. The wheel shall withstand without failure for one (1) minute, the limit side load specified in Table I.

4.3.3 Rubber Physical Properties Test: Unless otherwise specified below, rubber specimens taken from the tire shall be tested for Physical Properties in accordance with Specification FED-STD-601.

4.4.4 Rubber Hardness: The rubber hardness shall be determined by means of the Shore Durometer, Type A, or any Durometer similar and equal. The outer shell rubber shall register 70 ± 3 durometer. After all other testing specified herein is completed, a section shall be cut from the inner core of one (1) tire. This sample shall register 55 ± 3 durometer. After accelerated aging for seven (7) days in air at 158 ± 2°F, the hardness shall not differ from the original value noted by more than 5 points.

4.4.5 Adhesion: To determine the adhesion of the rubber to the wheel, the tire shall be cut transversely in the plane of the wheel axis down to the wheel rim. The end of the tire shall be gripped in the jaws of the testing machine and pulled from the wheel circumferentially in a direction normal to the adhered surface and at a rate of 2 inches per minute. The force necessary to perform this test shall be a minimum of 80 pounds per inch of tire width at 21°C (70°F). The bond strength shall exceed the tear strength of the rubber.

4.4.6 Compression Set: The compression set of the thread shall be measured as follows:

a. The outside diameter of the wheel tread shall be measured within 0.010 inch and the point of measurement shall be marked with a line not wider than 1/8 inch.

b. The wheel shall then be loaded to its normal rated load with the marked point of measurement at the point of ground contact and the tread deflection measured within 0.010 inch.

c. The loaded wheel shall be placed in an air oven and subjected to a temperature 69° to 70°C (158°F) for a period of 24 hours.

d. The loaded wheel shall be removed from the oven and allowed to cool at room temperature at approximately 21°C (70°F) for 30 minutes. The outside diameter of the wheel shall be measured at the marked point. The percentage of set shall not exceed 20 percent when calculated as follows:

$$C = \frac{(D-D1) \times 100}{R}$$

When: C = Percentage of set
D = Original diameter of wheel under (a) above
D = Final diameter of wheel under (d) above
R = Tread deflection under load (b) above

4.4.7 Elongation: Suitable tests shall be performed on rubber samples taken from the outer shell portion of the tire, to determine the elongation of the rubber. This value shall be not less than 300%.

4.4.8 Tensile Strength: The tensile strength as received shall be not less than 2000 psi on natural rubber and not less than 1500 psi on synthetic rubber.

4.4.9 Resistance to Abrasion: When tested as described in Specification FED-STD-601, the abrasion index shall be:

(1) Tire as received not less than 100%
(2) After accelerated aging as described in paragraph 4.4.4, not less than 80 percent of its original value

4.4.10 Roll Test: The sample wheel shall be pressed against a test wheel with an axial load equal to the rated load (Table I). The wheel shall be rolled for an equivalent of five (5) miles at a speed of 25 ± 2 miles per hour, rested for 15 minutes and then rolled an additional five (5) miles. The wheel shall then be rested for ten (10) hours and then rolled at the previously stated load and speed until failure occurs. Any failure at a distance less than an equivalent to 150 miles on this test shall be reason for rejection.

4.4.11 Overload Test: The sample wheel shall be pressed against the test wheel with a force equal to 1.5 times the rated load listed in Table I and rolled at a speed of 25 ± 2 miles per hour. The wheel shall operate in a satisfactory manner for a minimum of 30 minutes.

4.4.12 Towing Tests: The following tests are required.

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4.4.12.1 Towing Test Equipment: For the performance of this test, a suitable test trailer shall be provided by the contractor. The test trailer shall be of four wheel design, having front wheels steerable by means of tie-rods, suitably attached to a tow bar. The wheel base of the trailer shall be 7 ± 1 feet with a track tread of 5 ± 1 feet. The trailer shall include means of attachment of leads as required for the performance of the tests specified herein, and shall be capable of performing these tests without excessive deformation. The contractor shall provide as part of the test trailer, weights of the necessary number and sizes to load the test wheel as specified herein. The test trailer shall include means of mounting the test wheel in a front steerable position. The additional wheels required for the trailer need not be of the Model being tested but must be capable of satisfactory operation for the duration of the specified tests.

4.4.12.2 Towing Test Procedure: Under rated load and at an average of 10 miles per hour, the wheel shall be towed over the following courses:

a. Smooth paved highways for a distance of 100 miles
b. Graded gravel roads for 50 miles

The courses shall contain obstacles such as rail-crossings and door sills. At least fifty (50) short radius left-hand turns and twenty-five (25) sudden stops from full speed shall be accomplished during the test. At the conclusion of the tests, the wheel shall be examined for excessive wear or other damage. Excessive wear or damage to the wheel shall be cause for rejection. With an overload of 10% on the test wheel, the test trailer shall be towed over smooth concrete pavement in a circular path having a 35 foot radius as measured to the outside wheel of the trailer. The trailer shall make 5 complete circles counter-clockwise at a speed of 15 miles per hour. At the conclusion of the test the wheel and hub portion shall be inspected for permanent deformation or other damage. Permanent deformation or other damage shall be cause for rejection. Damage shall be considered over and above normal wear. Wear is considered excessive when flat spots, torn outer rubber shell, or splits over ½ inch long appear.

4.4.12.3 Speed Test: The wheel under the rated load shall be towed at an average speed of 20 miles per hour for a period of 15 minutes over smooth paved concrete. At the conclusion of this period, the wheel shall be allowed to stand for a period of 60 minutes, this cycle shall be repeated ten times. At the conclusion of the entire test, the wheel shall show no signs of excessive wear or other damage, nor shall the tire show any signs of separation from the wheel. Excessive wear, other damage, or tire separation shall be cause for rejection.

4.4.13 Environmental Tests: Environmental tests shall be conducted in accordance with the specified procedures of Specification MIL-E-5272. If the axle is removed, the grease seal bore shall be covered with a suitable closure. The sample wheel not destroyed by the previous test shall be used.

4.4.13.1 High Temperature Test: This test shall be conducted in accordance with procedure I, except the temperature shall be 125° F. At the conclusion of this test and with the temperature maintained, the wheel shall meet the test specified in 4.3.1 but with twice the load specified in Table I.

4.4.13.2 Low Temperature Test: This test shall be conducted in accordance with procedure I. At the conclusion of the test and with the temperature maintained, the wheel shall be subjected 20 times to an instantaneously applied load equal to the rated load specified in Table I, distributed along an axle 4 feet between wheel tread centers. The load shall be dropped a distance of 3 inches. No damage, distortion or tire separation shall result from this test.

4.4.13.3 Humidity Test: The test shall be conducted in accordance with procedure I, except the temperature shall not exceed 125 ± 5°F. At the conclusion of the test there shall be no evidence of corrosion or failure of the coatings.

4.4.13.4 Sand and Dust Test: This test shall be conducted in accordance with procedure I. During the test the wheel shall be rotated at a simulated road speed of 10-15 miles per hour. There shall be no evidence of dust particles in the bearing area and no substantial failure of the protective coatings or finishes.

4.4.13.5 Salt Spray Test: This test shall be conducted in accordance with the procedure specified for a period of 50 hours. There shall be no evidence of corrosion or failure of finishes and protective coatings.

4.5 Loading Table: The following table lists the Model, size, rated load and the side load for the wheel produced under this specification:

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TABLE I

Model	Wheel Size	Rated Load	Side Load
I	16 X 4	1600 lbs.	500 lbs.
II	21 X 6	3200 lbs.	1000 lbs.

5. PREPARATION FOR DELIVERY

5.1 Domestic Shipment for Immediate Use: Wheel assemblies shall be prepared for shipment in a manner which will insure arrival at destination in satisfactory condition at the lowest rate of the carrier.

5.2 Domestic Shipment Involving Storage:

5.2.1 Preservation and Packaging: Wheel assemblies shall be preserved and packaged as specified herein for overseas shipment.

5.2.2 Packing: Packaged wheel assemblies shall be packed as specified for overseas shipment, except the containers shall conform to Specification PPP-B-601.

5.3 Overseas Shipment:

5.3.1 Preservation and Packaging: Wheel assemblies shall be wrapped or bagged in barrier material conforming to Specification MIL-B-121, Grade A.

5.3.2 Packing: Unless otherwise specified by the procuring agency, each shipping container shall contain one unit (wheel assembly), it shall be uniform in size and shall be designed to enclose the contents in a snug, tight-fitting stackable container. Containers shall conform to Specification PPP-B-621. Plywood when used shall be Type II, Class 2.

5.4 Marking of Shipments: Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES:

6.1 Intended Use: Wheel assemblies covered by this specification are intended for use as standard components of mobile equipment used by the Armed Services under all extremes of climatic and environmental conditions.

6.2 Ordering Data: The procuring agency shall specify the following requirements:

a. Model wheel assembly desired
b. Number of assemblies per package
c. Desired method of preservation and packing

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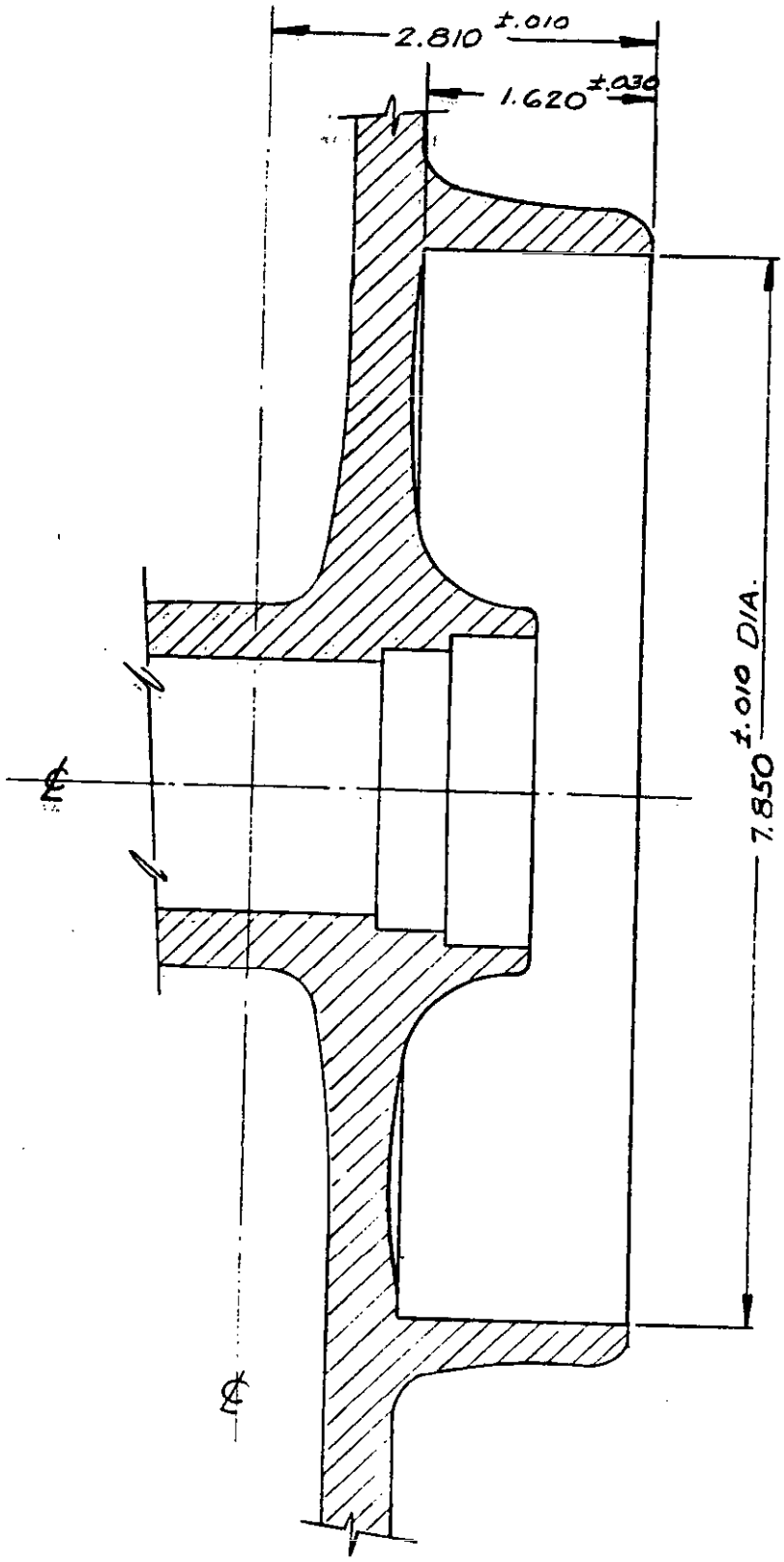


FIGURE 1

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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